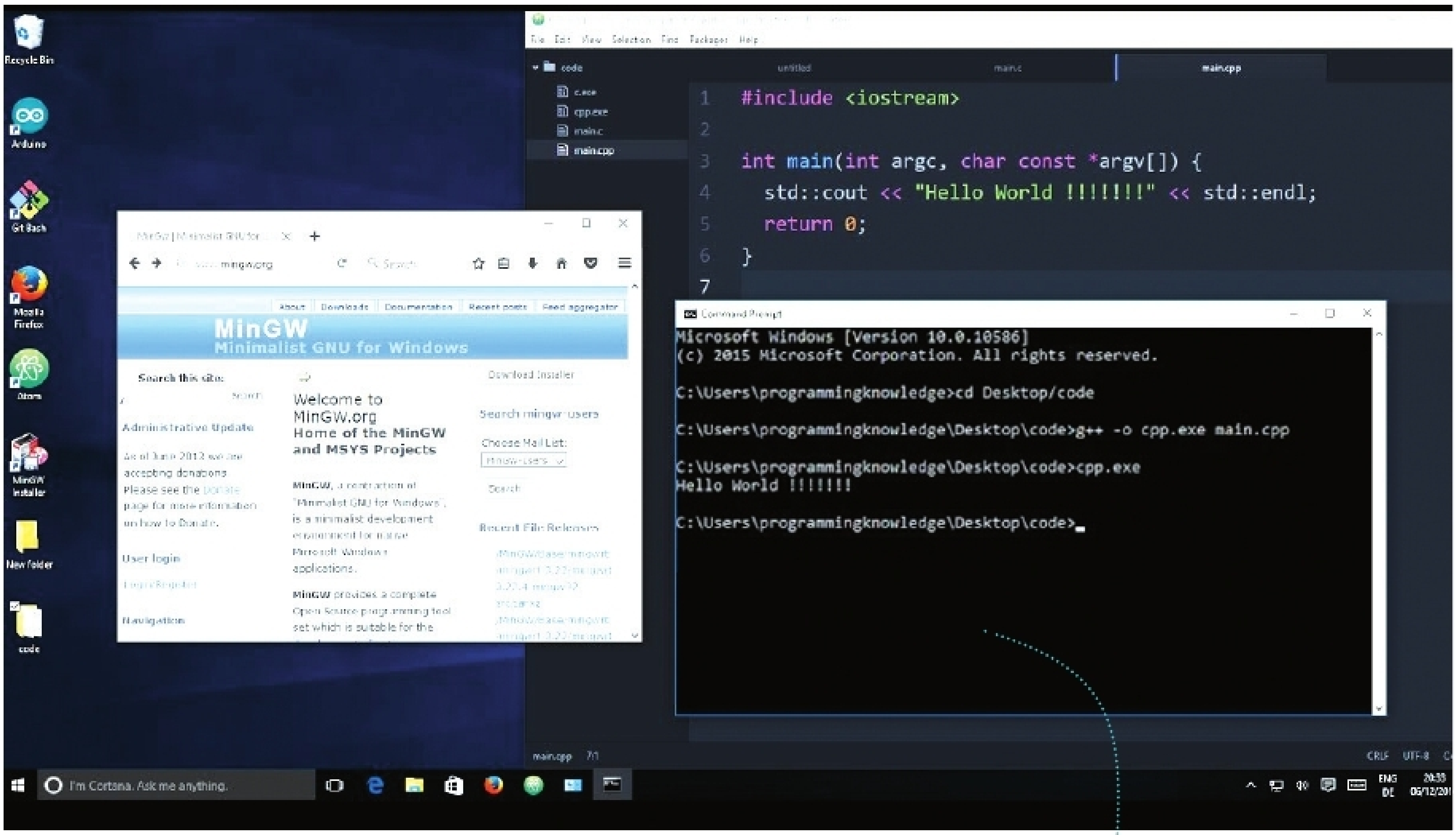

Using MinGW for C Programming in Windows

Though Windows and Linux are often perceived as incompatible with each other, it is possible to run free and open source software on Windows. C is still a very popular programming language. This article discusses how MinGW, an open source software development environment, can aid C programming in Windows.

Instead of thinking of open source alternatives to Microsoft Windows, it also makes sense to think about open source software for Microsoft Windows. According to Wikipedia, "In the area of desktop and laptop computers, the share of Microsoft Windows is generally above 70 per cent in most markets and at 78 per cent, globally." The Windows operating system is used both for technical and non-technical purposes. This article will focus on programming with C in Windows. C programming is still very popular, ranking among the top ten in almost all the programming language rankings. Since both C and Windows are still very popular, it is absolutely essential to learn about open source tools that can be used for programming in C in Windows.

The C language and UNIX have a very close-knit relationship. Any article on UNIX will mention C programming in the first paragraph itself, and vice versa. Some of the header files in C are developed specifically for UNIX. For example, header files like *unistd.h*, *signal.h*, etc, are added as part of the POSIX standard for UNIX-specific functionalities. Please note that whatever applies to UNIX here is also applicable to Linux. So C programming being closely related to UNIX makes it slightly troublesome when used on non-UNIX operating systems. Since MacOS is based on BSD Linux, it is relatively easy to do C programming in it. But programming with C in Windows is more problematic.

Let us discuss the different ways to program with C in Windows. There are still people who use Turbo C++ or Borland C++ on older versions of Windows. The use of such outdated IDEs is not a good practice but since this is often restricted to academic institutions, it need not be taken seriously. Serious C programmers on Windows typically use Microsoft Visual